

A long-form single-speaker real-time MRI speech dataset and benchmark

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Motivation

- real-time (rt)-MRI bestows the most in-depth view of the moving vocal tract during speech production
- rtMRI data has numerous applications, e.g. articulatory synthesis, speech inversion, articulatory representation learning
- previous publicly datasets have been limited in the amount of single-speaker data, limiting the use of such data for various speech processing tasks

Dataset	Read (min)	Spont. (Min)	FPS
USC TIMIT	~37	0	23
USC 75 Speaker	~12	~5	86
USC LSS	37	17	99

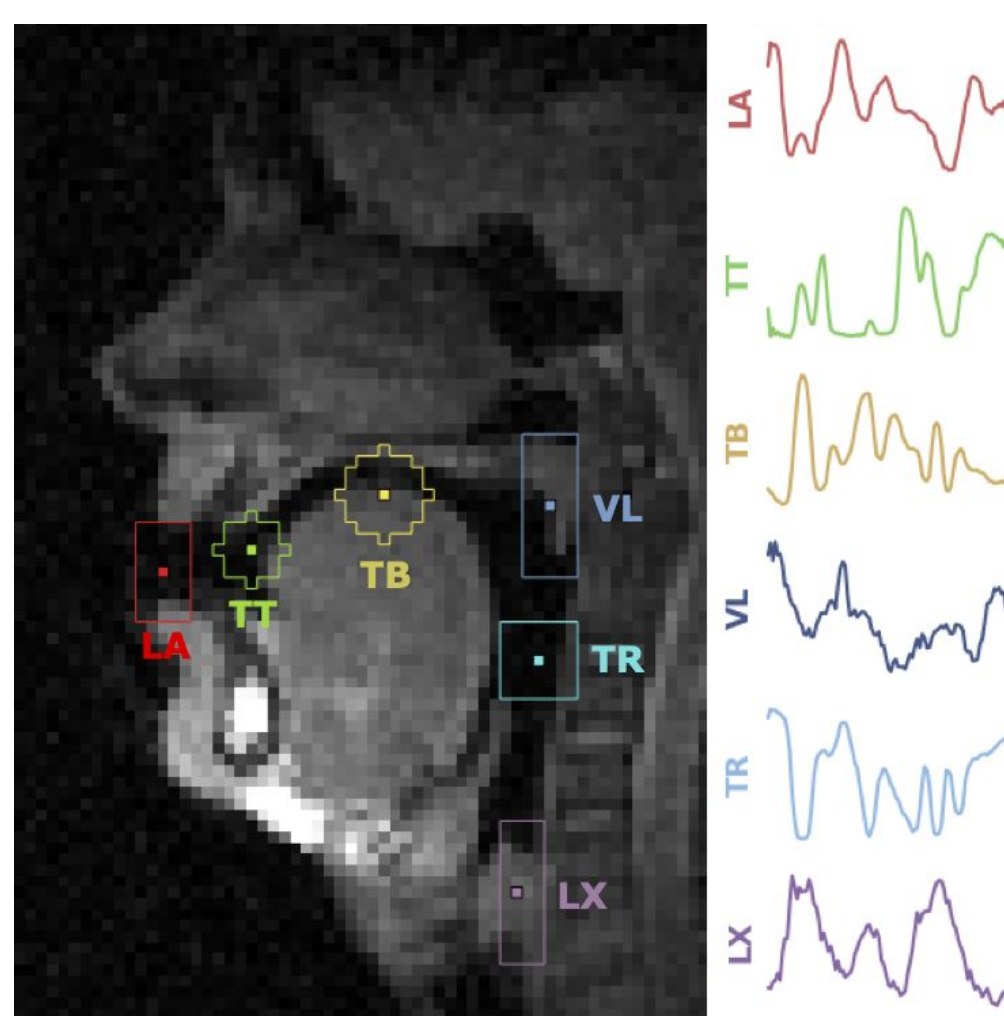
USC LSS Dataset

- **Data acquisition:** 0.55T MRI scanner; 99 FPS video; 16kHz audio; 71 scans ~ 54 min of speech
- **Stimuli:** USC TIMIT sentences, various passages, spontaneous speech
- **Subject:** male, age 32, native speaker of American English
- **Audio derivations:** denoised and restored audio
- **Sentence-level split:** 684 sentences; train/val/test splits
- **Region-of-interest (ROI) timeseries:** ROIs for six crucial vocal taction regions track articulator kinematics (3)



(1)

(2)

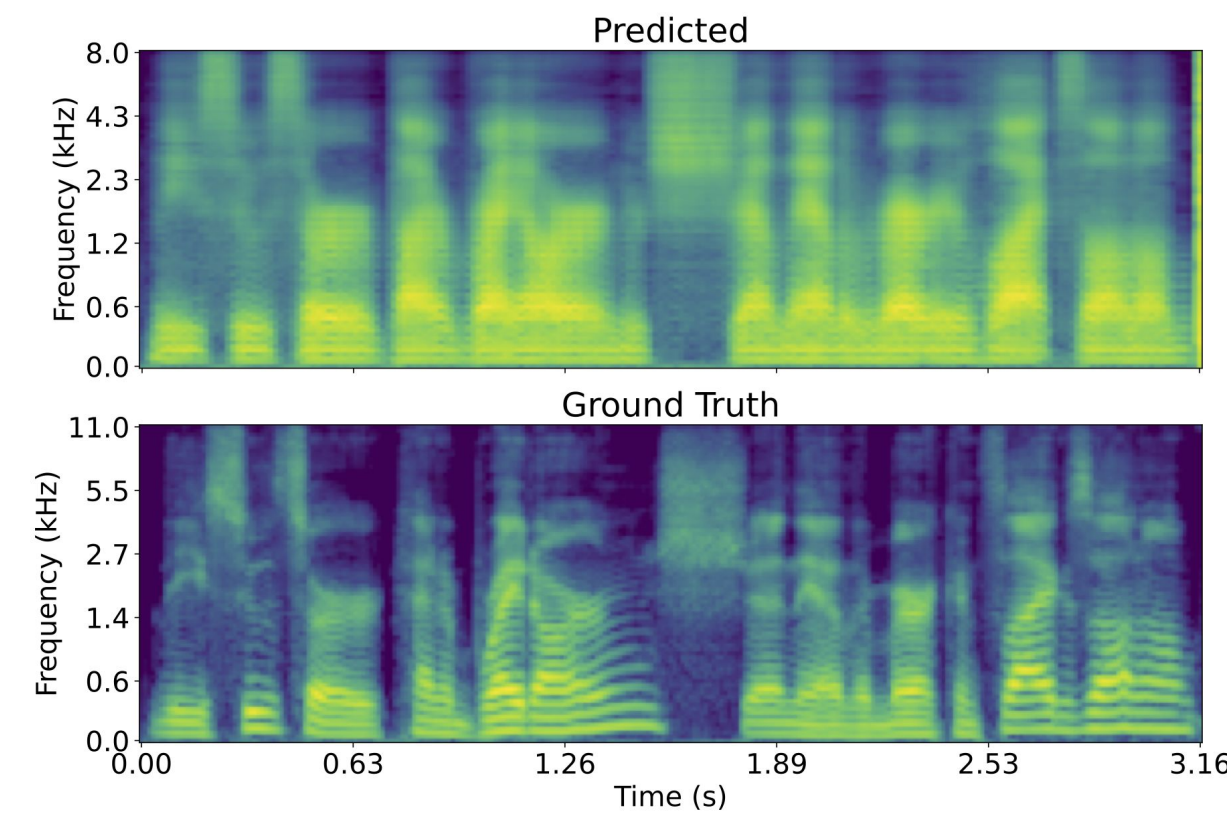


(3)

Benchmarks

Articulatory Synthesis

- **Model:** HiFi-GAN (last 4 G blocks & D frozen)
- **Input:** denoised & restored audio; resized video [H x W, t]
- Batch size = 2; LR = 1e-3



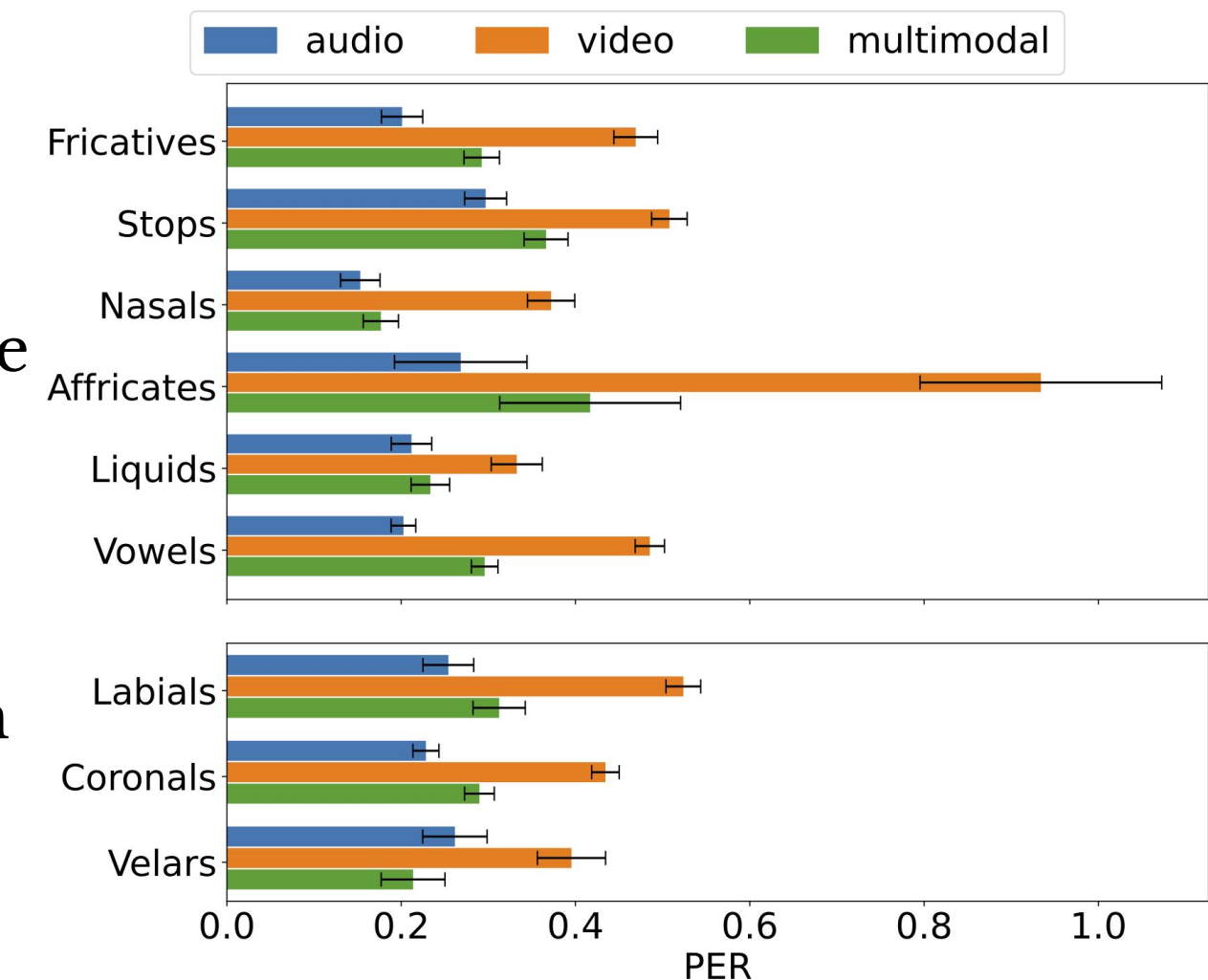
(4)

Audio	CER (%) ↓	WER (%) ↓	MCD ↓
Denoised	31.3 ± 6.0	52.6 ± 10.1	4.3 ± 0.1
Restored	24.3 ± 4.6	45.0 ± 8.2	4.8 ± 0.2

(5)

Phoneme Recognition

- **Model:** Conformer + LSTM
- **Baseline:** Wav2vec2phoneme
- **Input:** unimodal audio or video; multimodal audio + video
- **Features:** audio = latents from 9th layer of WavLM; video = CLS token from fine-tuned ViT
- **Loss:** CTC loss; batch size= 16; LR = 1e-4



(6)

Model	PER ↓
Wav2Vec2Phoneme	0.39 ± 0.03
Audio	0.22 ± 0.03
Video	0.50 ± 0.02
Multimodal	0.30 ± 0.03

(7)

Applications

1. Articulatory synthesis & speech inversion

Previous work has been limited by availability of long single-speaker data and cross-speaker learning is challenging.

2. Articulatory representation learning

Self-supervised learning directly from rtMRI data has potential applications to model cognitive aspects of speech production, use in BCI, and speech biofeedback tools.

3. Improvement of vocal tract segmentation methods

Vocal tract contours allow for deriving representations that capture only speech-relevant information, though methods for extracting such contours from 99 FPS video are underdeveloped.

4. Phonetic analyses

Potential for studying within-speaker variation and uniformity in articulatory implementation.